

NEURAL BASIS OF LANGUAGE & SPEECH

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Lecturer

DIKIOHS, DUHS.

LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- Describe the neural basis of language & how the brain organizes language functions.
- Identify the major cortical areas involved in language comprehension & speech production including Broca's & Wernicke's area.
- Explain the role of dominant hemisphere in language processing.
- Describe the role of Broca's area, Wernicke's area, arcuate fasciculus, motor cortex & related cortical regions in language & speech.
- Define aphasia & differentiate between Broca's & Wernicke's aphasia.
- Relate types of aphasia to the underlying lesions.

What is language?

Language is a system used to:

- Express thoughts & ideas
- Understand spoken & written information
- Communicate emotions & intentions
- Store & retrieve information
- Interact socially

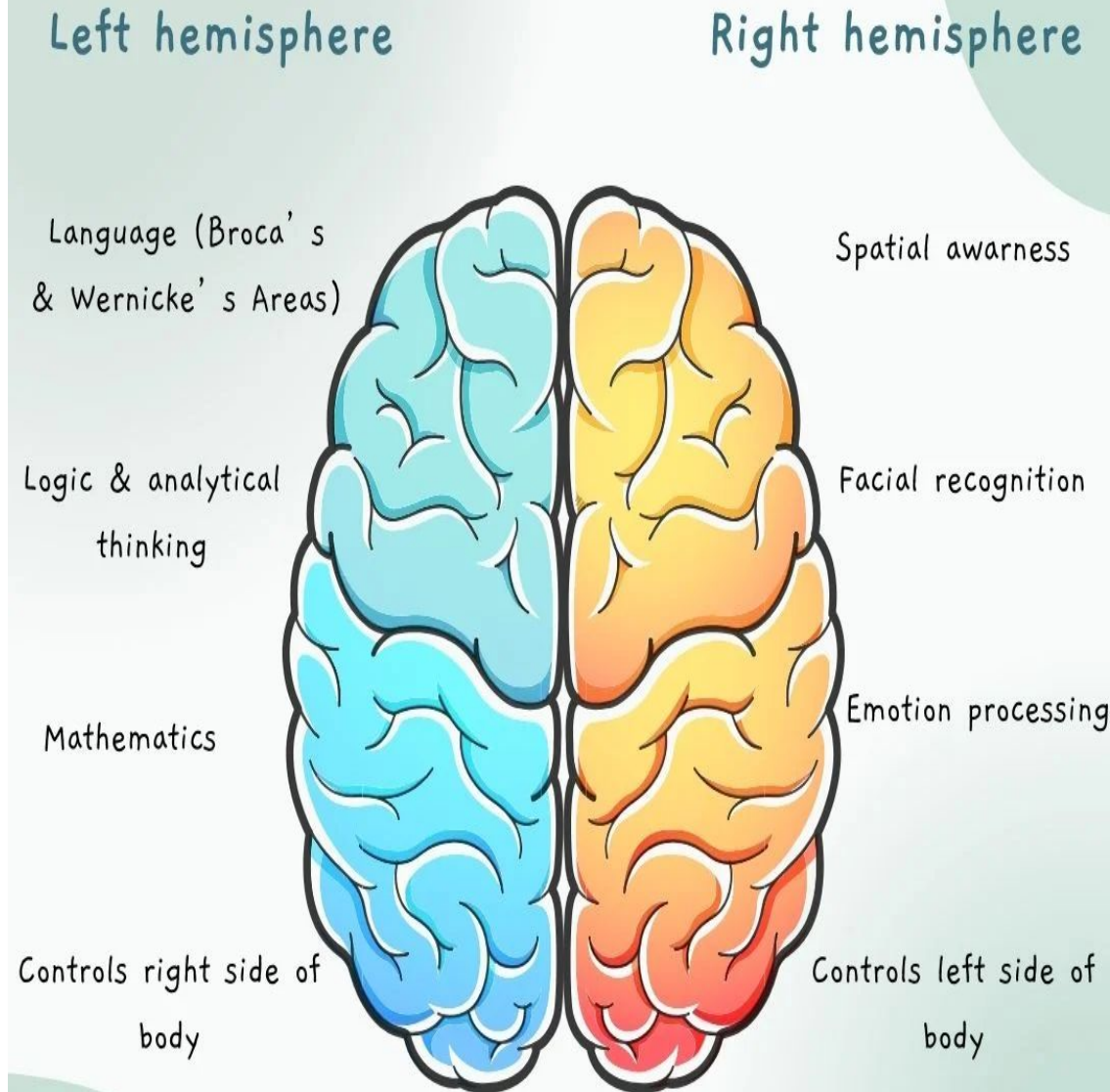
There are two main components of language:

1. Language comprehension
2. Language production

ORGANIZATION OF LANGUAGE IN THE BRAIN

DOMINANT CEREBRAL HEMISPHERE

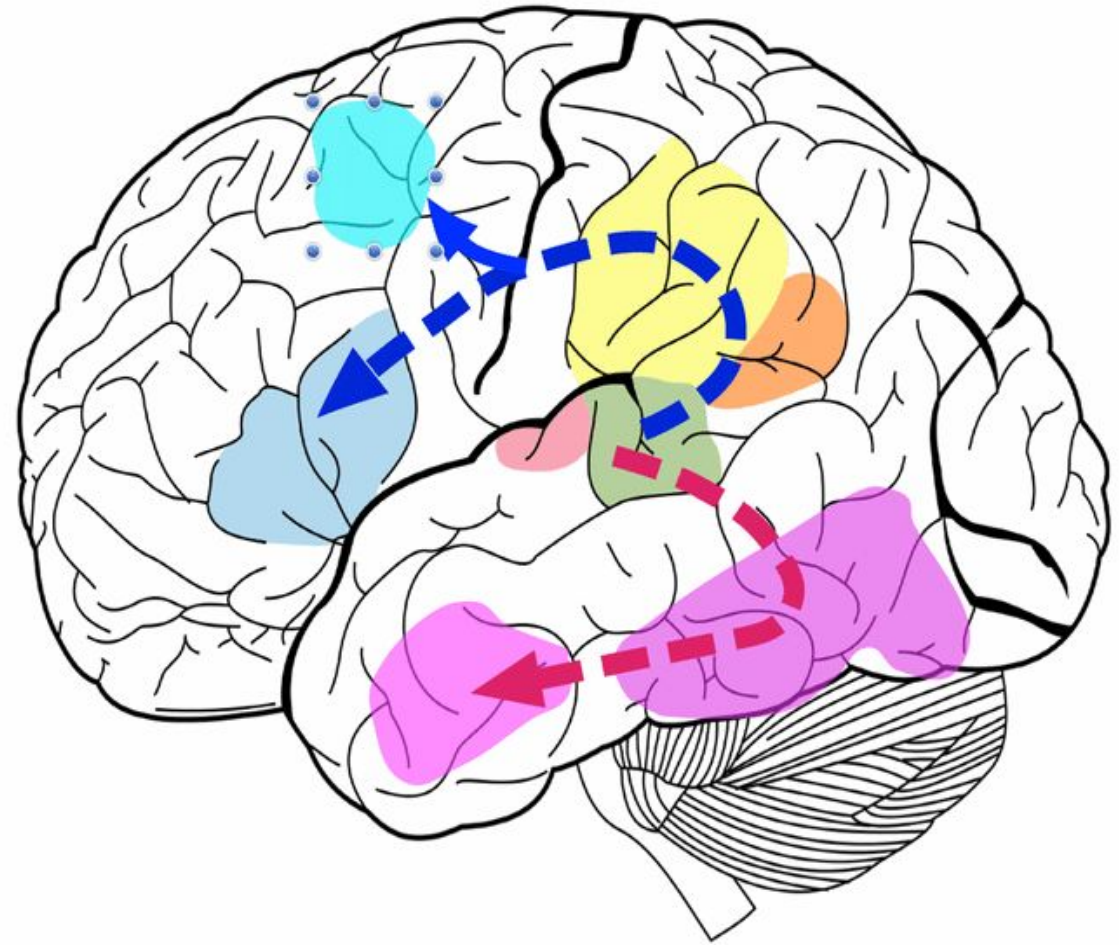
- The left cerebral hemisphere is the dominant hemisphere for language in approximately 90-95% of right-handed individuals.
- Language dominance can vary in left-handed individuals.

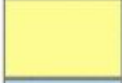
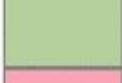


IMPORTANT LANGUAGE AREAS

□ Important cerebral areas for language are:

1. Broca's area
2. Wernicke's area
3. Primary auditory cortex
4. Primary motor cortex
5. Angular gyrus
6. Supramarginal gyrus
7. Arcuate fasciculus
8. Supplementary/premotor areas
9. Prefrontal cortex



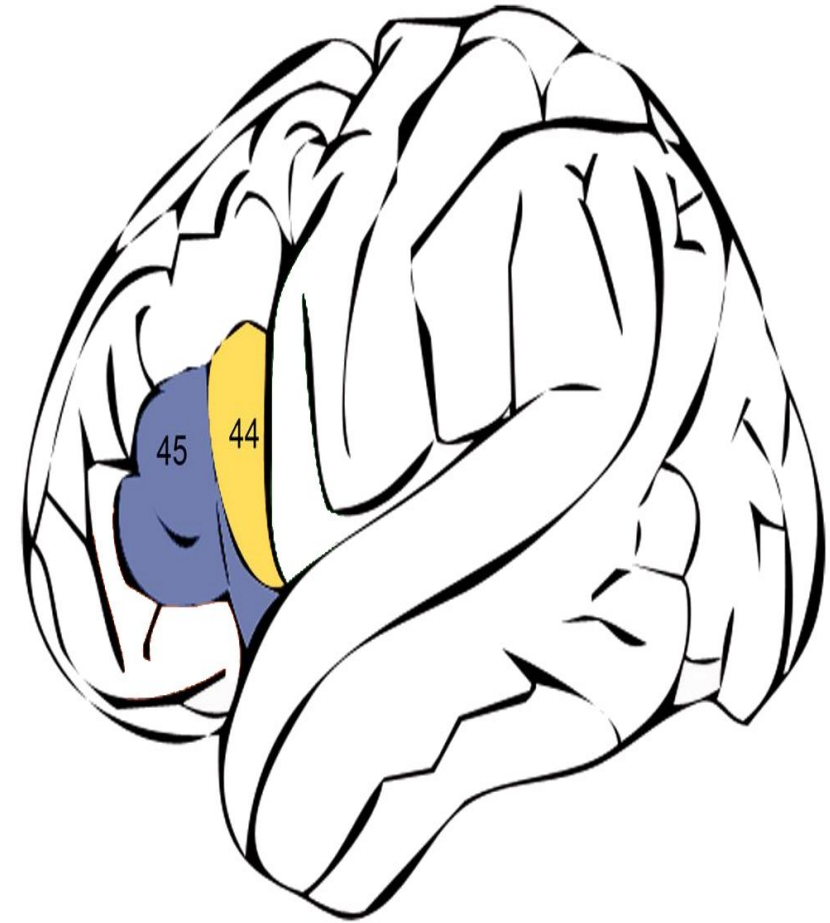
	Angular Gyrus
	Supramarginal Gyrus
	Broca's Area
	Wernicke's Area
	Primary Auditory Cortex

BROCA'S AREA

- It is located in the inferior frontal gyrus & is composed of Brodmann area 44 & Brodmann area 45
- It is usually located in the left hemisphere for vast majority of right-handed individuals & many left-handed individuals.

FUNCTIONS

- Motor planning of speech
- Organization of speech output
- Articulation
- Sentence formation
- Grammar & syntax processing
- Conversion of linguistic information into a motor speech program

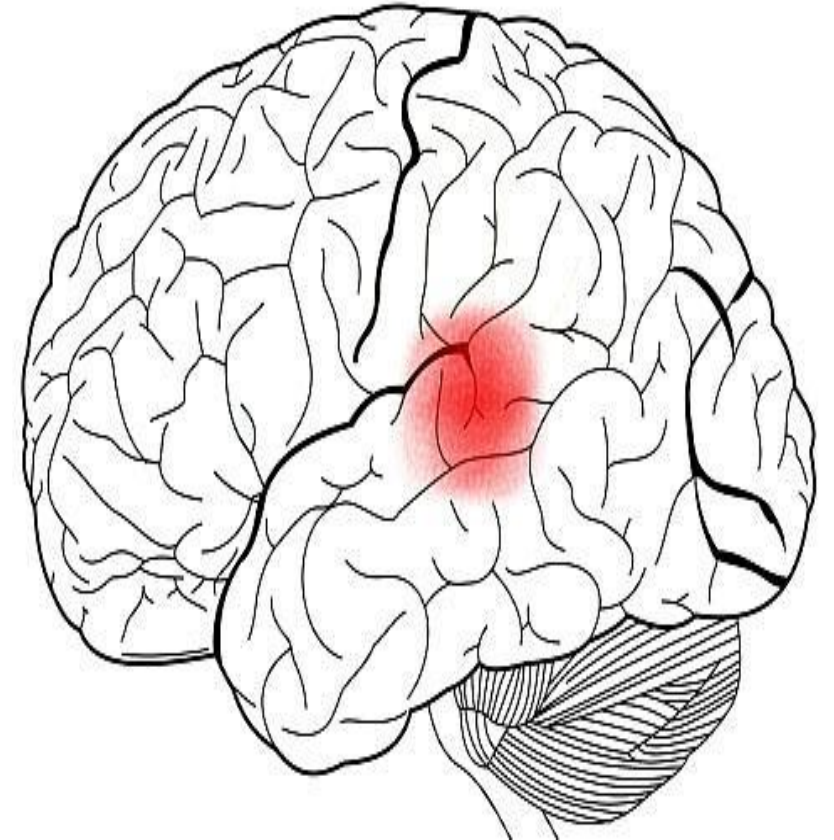


WERNICKE'S AREA

- It is located in the posterior part of upper temporal gyrus of dominant hemisphere.
- It corresponds to Brodmann area 22 & is essential for understanding the spoken & written language

FUNCTIONS

- Understanding spoken language
- Recognition of meaningful words
- Interpretation of language
- Semantic processing
- Association of words with their meaning



PRIMARY AUDITORY CORTEX

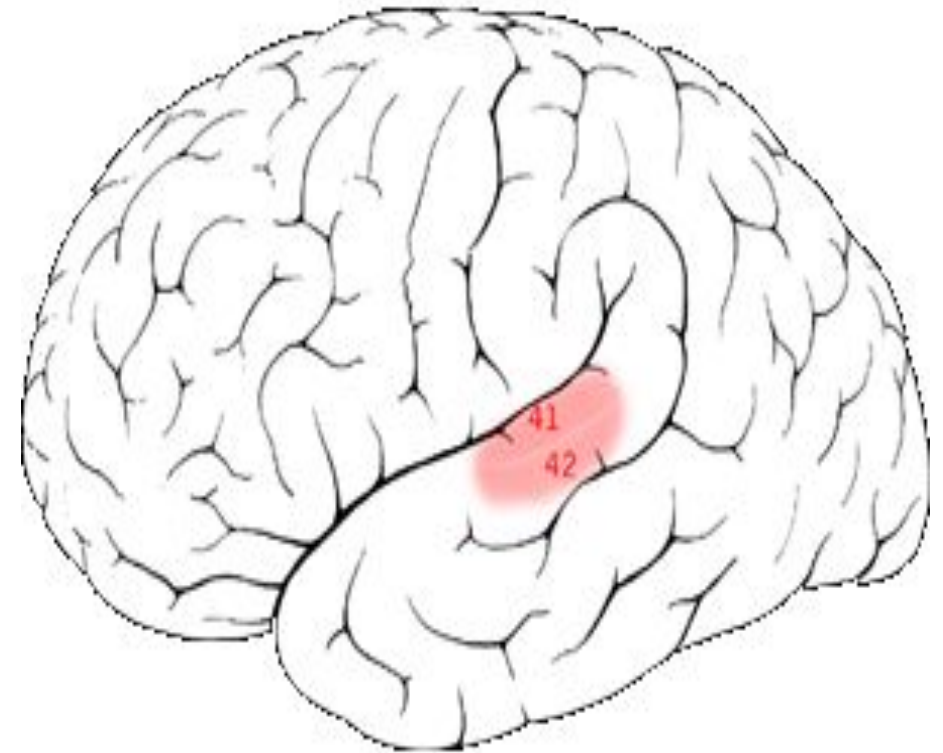
- It is located on the transverse temporal gyri of Heschl in the superior part of temporal lobe.
- It corresponds to Brodmann area 41 & 42.

FUNCTIONS

- Receives & analyzes auditory information
- Detects frequency, intensity & pattern of sounds

Sound -> auditory cortex -> language processing areas.

Primary auditory cortex



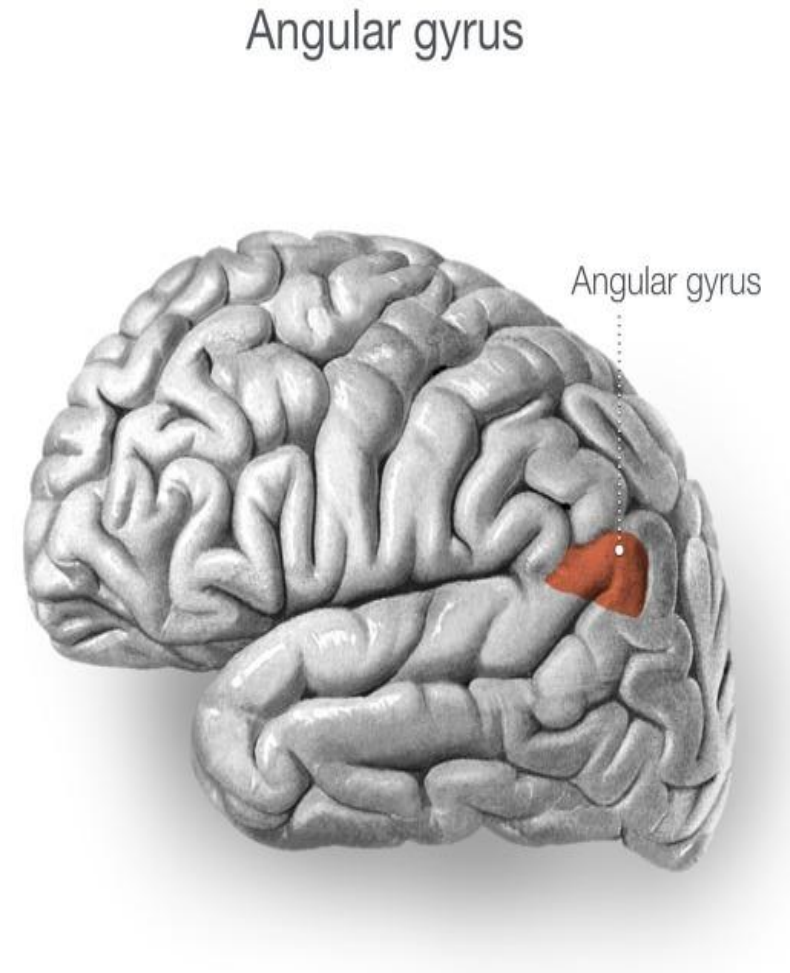
ANGULAR GYRUS

- It is located in the inferior part of parietal lobe next to occipital & temporal lobes.
- It corresponds to Brodmann area 39

FUNCTIONS

- Links visual inputs(e.g. reading a book) with auditory processing areas like Wernicke's area to comprehend information.
- Translates written letters into recognizable sounds & meaningful concepts.

Visual word -> angular gyrus -> language network.



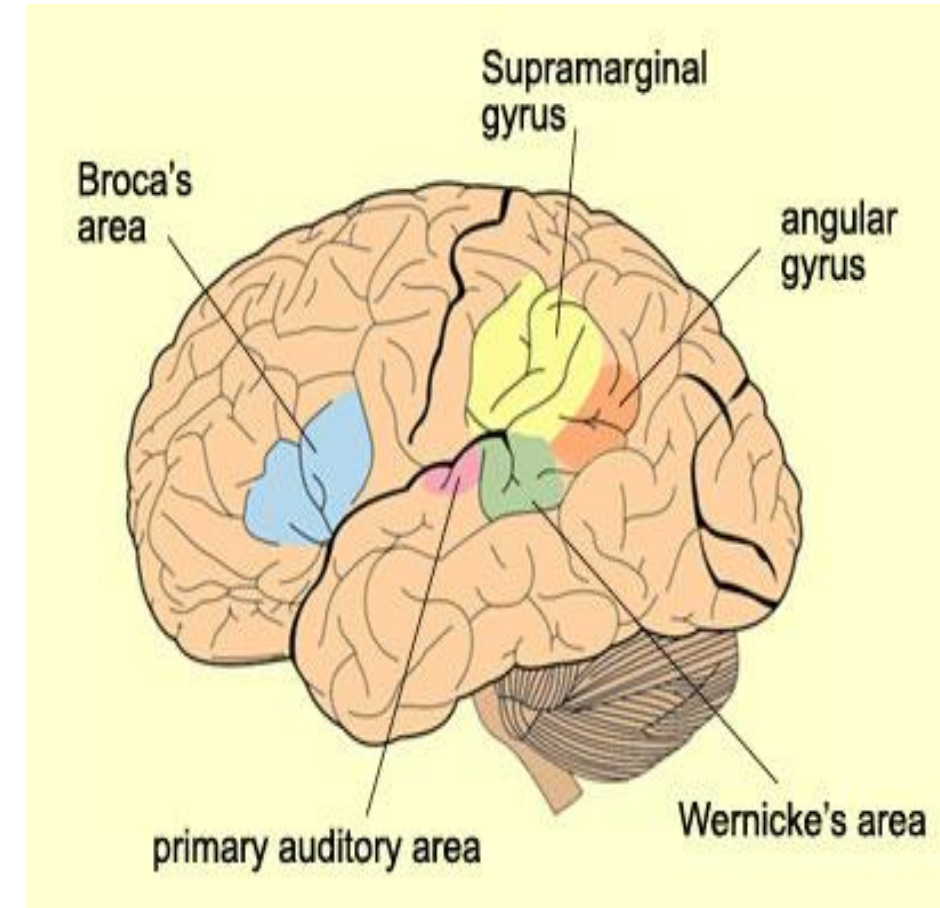
SUPRAMARGINAL GYRUS

- It is located in the inferior parietal lobule of the parietal lobe.
- It corresponds to Brodmann area 40.

FUNCTIONS

- It is associated with phonological processing of spoken & written language.
- Manages verbal working memory & involved in integration of auditory & articulatory information.
- It helps the brain:

Hear a word -> hold its pattern -> reproduce it.

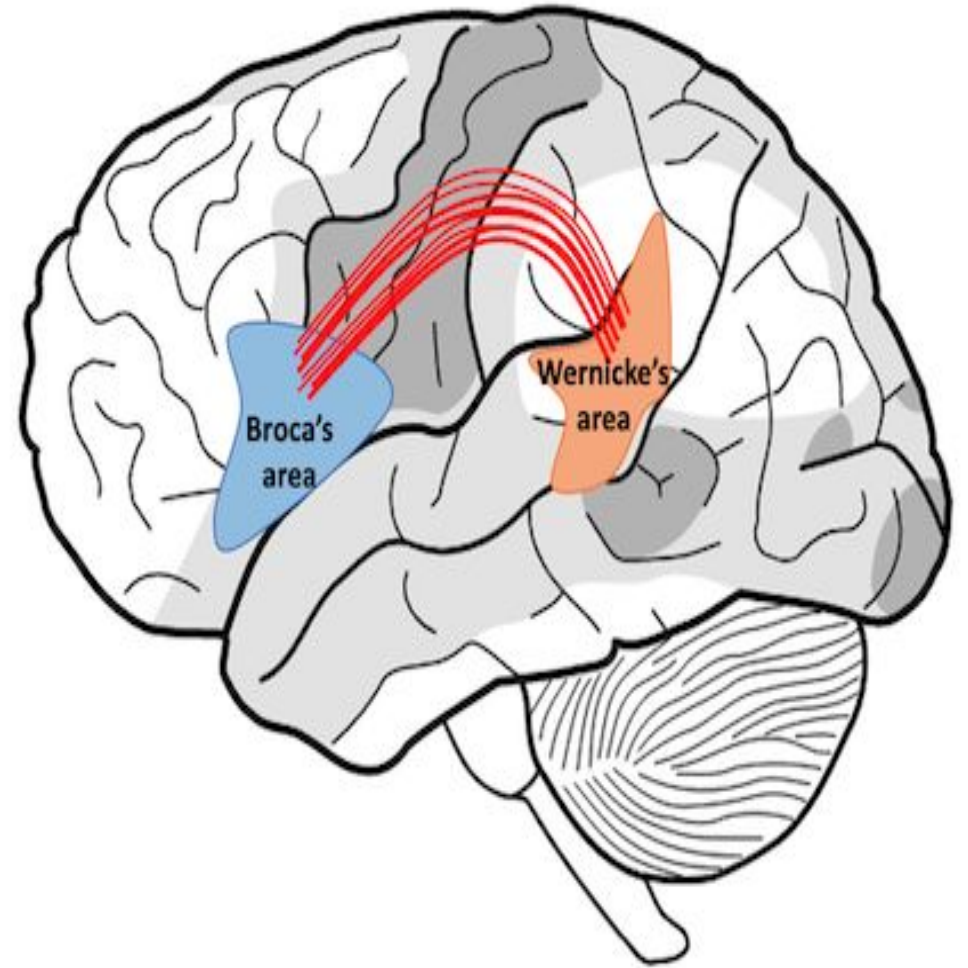


ARCUATE FASCICULUS

- It is a major white matter pathway that connects posterior temporal language regions to frontal language regions.
- It allows communication between language comprehension system & speech production system.

FUNCTION

- Particularly important for repetition & phonological processing of words.
- Involved in transfer of linguistic information from comprehension to production.



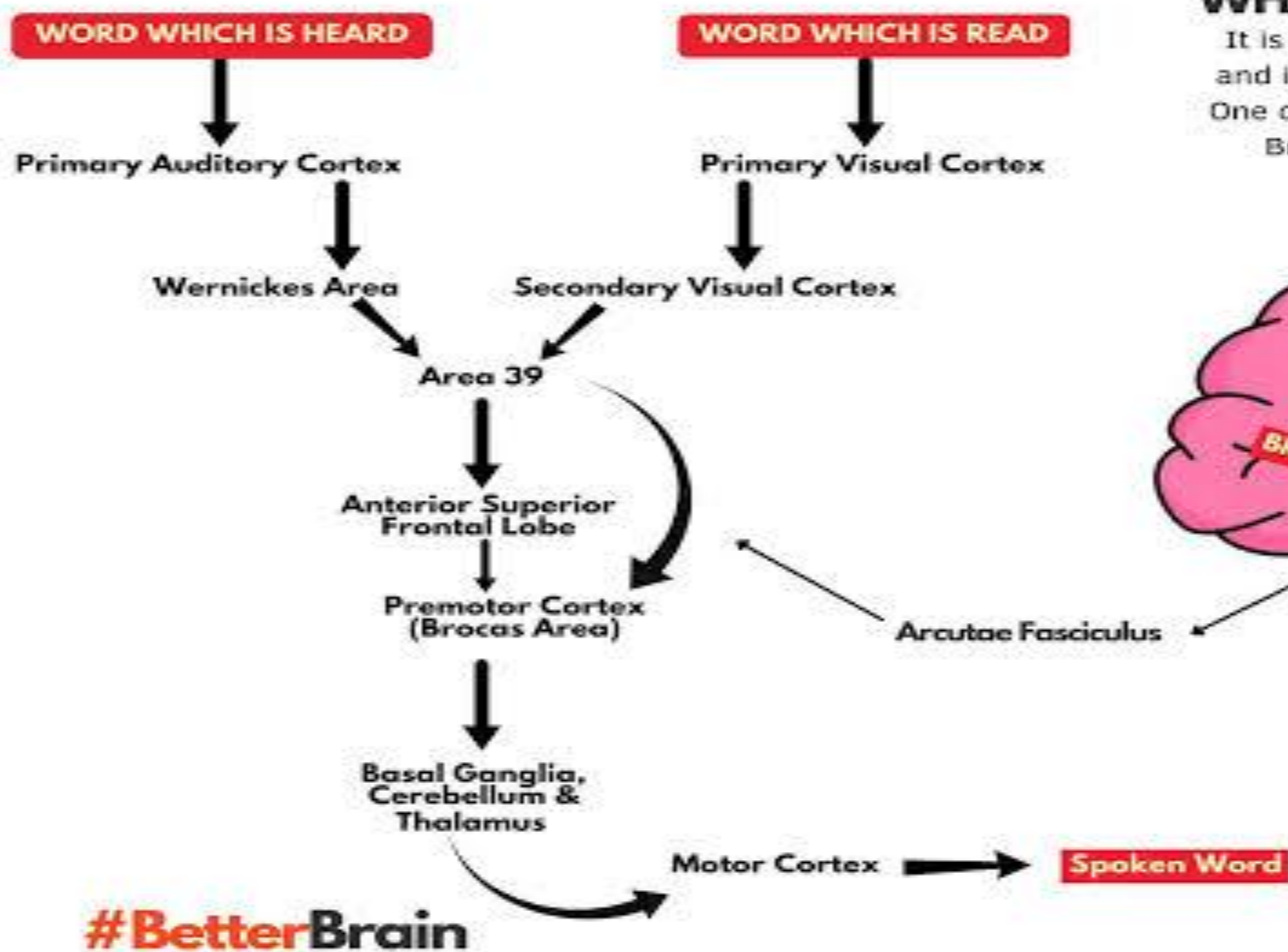
MECHANISM OF LANGUAGE COMPREHENSION

□ Neural pathway from hearing to speaking includes:

Auditory input -> primary auditory cortex -> Wernicke's area -> arcuate fasciculus -> Broca's area -> premotor/motor cortex -> corticobulbar pathway -> cranial nerve nuclei -> speech muscles

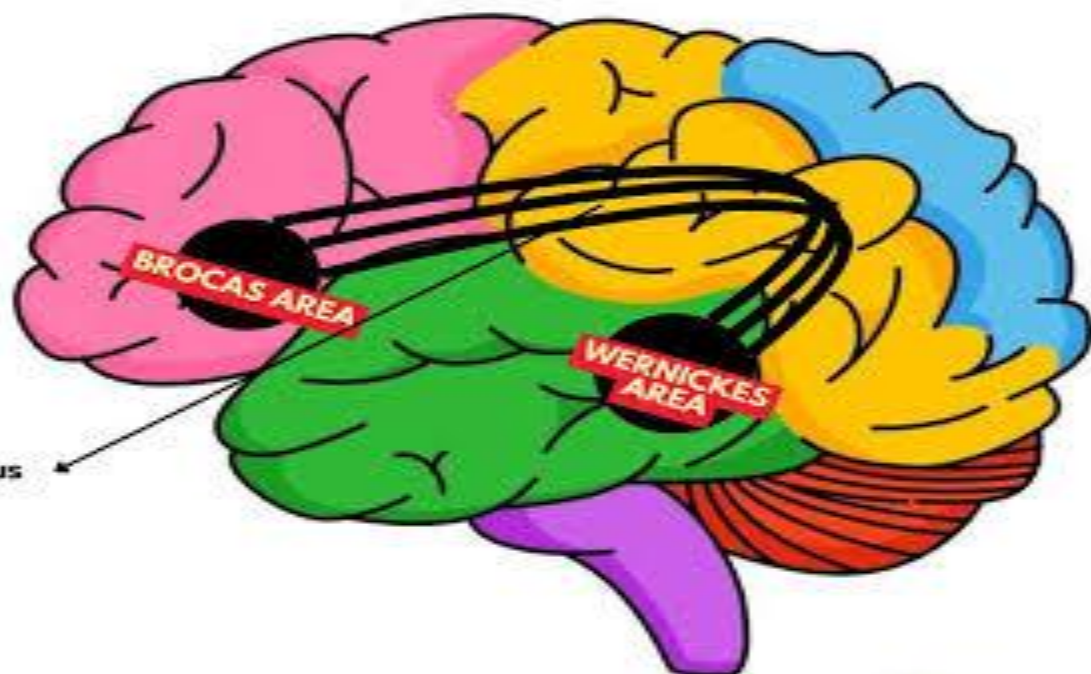
1. Sound waves enter the ear
2. Auditory receptors convert sound into neural signals
3. Signals travel through the auditory pathway
4. Primary auditory cortex analyzes the sound
5. Language areas identify speech sounds
6. Words are recognized & meaning is assigned
7. Sentence meaning is integrated
8. Appropriate response is formulated.

LANGUAGE PATHWAYS IN THE BRAIN



WHAT IS THE ARCUATE FASCICULUS?

It is a bundle of axons that connects the temporal cortex and inferior parietal cortex to locations in the frontal lobe. One of the key roles of the arcuate fasciculus is connecting Broca's and Wernicke's areas, which are involved in producing and understanding language.



What is speech?

□ Speech is defined as:

The expression of thoughts by production of articulate sounds bearing a definite meaning.

□ When a sound is produced verbally, it is called speech.

□ If it is expressed by visual symbols, it is termed as **writing** & if visual symbols are expressed verbally, it is called **reading**.

□ Speech depends upon coordinated activities of **central speech apparatus** i.e. the cortical & subcortical areas & **peripheral speech apparatus** i.e. mouth, lips, tongue, pharynx & larynx.

□ Peripheral structures involved in speech work in coordination with respiratory system under the influence of motor impulses from respective motor areas of cerebral cortex.

ROLE OF CORTICAL AREAS IN THE DEVELOPMENT OF SPEECH

□ Development of speech involves integration of three cortical areas i.e.

1. Wernicke's area
2. Broca's area
3. Motor area

WERNICKE'S AREA-SPEECH COMPREHENSION

- Understanding of speech starts in Wernicke's area located in the temporal lobe of cerebral cortex.
- It is responsible for understanding the visual & auditory information required for speech.
- After understanding the words it sends information to Broca's area.
- It sends fibers to Broca's area through arcuate fasciculus.

BROCA'S AREA-SPEECH SYNTHESIS

- Speech is formulated in Broca's area located adjacent to the motor area that drives the movements of tongue, lips, larynx necessary for speech.
- It receives information required for the production of words from Wernicke's area & develops pattern of motor activities required for speech.
- The formulated pattern of motor activities is then sent to the motor area.

MOTOR AREA-ACTIVATION OF PERIPHERAL SPEECH APPARATUS

- It receives pattern of motor activities from Broca's area & activates the peripheral speech apparatus to initiate movements required for speech.

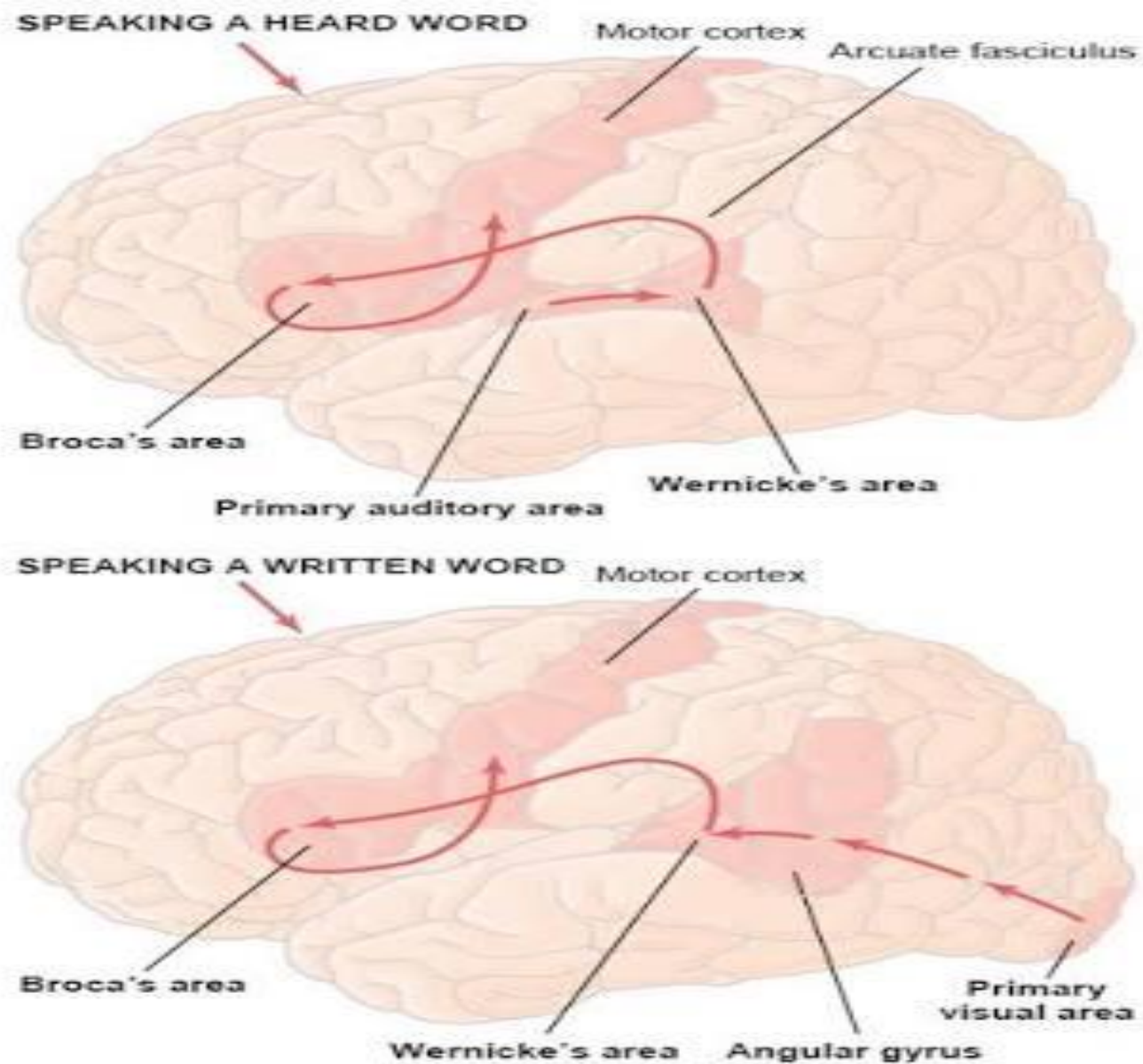


Figure 57-8

Brain pathways for (top) perceiving a heard word and then speaking the same word, and (bottom) perceiving a written word and then speaking the same word. (Redrawn from Geschwind N: Specializations of the human brain. *Sci Am* 241:180, 1979. © 1979 by Scientific American, Inc. All rights reserved.)

MECHANISM OF SPEECH DEVELOPMENT

- First stage in the development of speech is the association of certain words with visual, tactile, auditory & other sensations aroused by objects in the external environment.
- Association of words with these sensations is stored as **memory**.
- When a definite meaning is attached to certain words, neural pathway between the **auditory area (Heschl area-area 41)** & **motor area** that drives muscles of articulation (**Broca's area-area 44**) is established.

STAGES OF SPEECH DEVELOPMENT

1. RESPIRATION

- Process begins in the lungs & provide pulmonic air needed to power speech.
- During speech, pattern of breathing shifts i.e. inhalation becomes short & exhalation becomes long & controlled.

2. PHONATION

- Air from the lungs travel up the trachea & passes through the larynx(voice box).
- Air pressure forces the vocal cords to vibrate, turning stream of air into a buzzing sound source.

3. RESONATION

- Raw sound from the larynx is amplified & modified as it travels through the vocal tract(i.e. the pharynx, oral cavity & nasal cavity).

4. ARTICULATION

- Structures in the mouth modify the airflow to shape the sound into recognizable words.

LANGUAGE VS SPEECH

LANGUAGE

- It is the entire system of codes, symbols & rules used to convey meaning.
- Not necessarily spoken but can be written.
- Mainly involves cortical language network.

SPEECH

- It is the oral or spoken type of communication.
- It is a motor skill that involves muscles movement in the mouth, throat & chest.
- Involves motor cortex, brainstem & cranial nerves.

What's the difference?

Speech vs Language

Speech

is how we
produce sounds
and say words.



Language

is how we
understand
words and use
words to
communicate
thoughts,
feelings and
ideas.

APPLIED PHYSIOLOGY

APHASIA

- It is a speech disorder that affects a person's ability to speak, understand, read & write.
- It is caused by **damage to the brain's language centers** due to stroke, head injury, cerebral tumors, brain infection.
- Symptoms include difficulty in finding the right words, understanding spoken or written language & trouble spelling & forming sentences.

TYPES OF APHASIA

BROCA'S APHASIA

- It is also known as **expressive aphasia** or **anterior aphasia** & is due to damage of left frontal lobe which is also responsible for motor activities.
- Often associated with weakness or paralysis of right arm or leg.
- The affected person often talks in small phrases because of inability to construct sentences

WERNICKE'S APHASIA

- It is also called **receptive aphasia** or **posterior aphasia** & is due to lesion in left temporal lobe.
- It is characterized by fluent speech but speech without meaning.
- The affected people speak long sentences without any meaning & use incorrect or non-existent words.

GLOBAL APHASIA

- It is the most common type of aphasia characterized by combined features of Broca's aphasia & Wernicke's aphasia.
- It is due to widespread lesion in speech areas caused by infarction of left cerebral hemisphere.
- The affected persons can neither speak nor understand the spoken words & cannot even read or write.

NOMINAL APHASIA

- It is also called **anomic aphasia** or **amnesic aphasia** & is due to lesion in posterior temporal & inferior parietal gyri.
- It is characterized by inability in naming familiar objects.

MOTOR APHASIA

- It is also called **verbal aphasia** or **dyspraxia** & is due to defect in the pathway between left speech areas & excitomotor(precentral) cortex.
- The affected people are able to decide what to talk but cannot pronounce all the words due to lack of coordination between central speech apparatus & peripheral speech apparatus.

SENSORY APHASIA

- It is characterized by inability to understand words or symbols & is of two types, **auditory aphasia** or **visual aphasia**.

Cont..

- **Auditory aphasia(word deafness)** is characterized by inability to understand the spoken words & is due to lesion in secondary auditory area.
- **Visual aphasia(word blindness)** is characterized by difficulty in reading as a result of inability to understand written symbols due to lesion in secondary visual area.

AGRAPHIA

- It is characterized by inability to write & is due to the defect in the connection between cortical areas concerned with writing.
- Agraphia differs from dysgraphia which is characterized by distorted writing or writing incorrect letters.

Common Types of Aphasia

Broca's Aphasia

Good comprehension,
poor expression

- AKA non-fluent aphasia
- Express themselves in single words or short sentences
- Can lead to frustration for both speaker and listener

Wernicke's Aphasia

Good expression,
poor comprehension

- AKA fluent aphasia
- Difficulty understanding spoken language
- Can lead to frustration for speaker when listener cannot comprehend

Anomic Aphasia

Fluent speech, difficulty
finding words

- AKA anomia
- Milder type of aphasia
- Often use vague terms or describe the function of objects

Global Aphasia

Impaired in all aspects
of language

- Most severe form of aphasia
- Damage to multiple language areas in the brain
- Both speaking and understanding language very difficult

DYSARTHRIA/ANARTHRIA

- Disturbed articulation is termed as **dysarthria**.
- Anarthria means inability to speak.
- So, dysarthria or anarthria can be defined as difficulty or inability to speak due to paralysis or ataxia of muscles involved in articulation.
- It is caused by damage to the brain or nerves that drive the muscles of articulation like stroke, brain injury, degenerative diseases like Parkinson or Huntington disease.

DYSPHONIA

- It is a voice disorder characterized by hoarseness, dry & sore throat.
- Hoarseness is the difficulty in producing sound while trying to change the pitch or loudness of sound.
- Common causes include trauma, paralysis or lumps of vocal folds, inflammation of larynx & hypothyroidism.

STAMMERING (SHUTTERING)

- It is a speech disorder caused by neurological incoordination of speech characterized by disturbed speech flow due to repetitions, prolongations or abnormal stoppage of sounds & syllables.
- Exact cause is unknown however it may be due to genetic factors, brain damage or anxiety.

What is 'stuttering'?

Stuttering is an interruption to the flow of speech. This can happen in different ways:

- Repeating sounds "c c c can I have a biscuit?"
- Repeating words "my my my name is John"
- Prolongations "ssssocks and shoes"
- Blocks "w... what is that?"

A person who stutters may also look like they are tense and struggling to speak.

POST ASSESSMENT

Q1. Which cortical region is important in processing written language & associating visual information with language?

- a) Angular gyrus
- b) Precentral gyrus
- c) Postcentral gyrus
- d) Insular cortex

Q2. A patient can hear sounds but has difficulty interpreting spoken words. Which cortical area is most likely affected?

- a) Broca's area
- b) Wernicke's area
- c) Primary motor area
- d) Premotor area

Cont..

Q3. Which structure is particularly important for connecting language comprehension with speech production?

- a) Internal capsule
- b) Corpus callosum
- c) Medial lemniscus
- d) Arcuate fasciculus

Q4. Which sequence best represents the general process involved in speech production?

- a) Motor execution -> language comprehension -> language formulation.
- b) Motor cortex -> Wernicke's area -> retina.
- c) Language formulation -> speech motor planning -> motor execution.
- d) Hearing -> spinal reflex -> speech.